Quantification: 543_3
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	10
interior lacuna count	9
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 9)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2187.22	2174	598.8	px^2
major_axis_length_px	lacuna shape	82.3704	87.6934	10.0353	px
minor_axis_length_px	lacuna shape	36.7931	35.2412	11.1805	px
aspect_ratio	lacuna shape	2.387	2.2947	0.6949	unitless
eccentricity	lacuna shape	0.8811	0.9	0.0775	unitless
solidity	lacuna shape	0.8856	0.9295	0.0959	unitless
roots_count	headline	6.6667	7	2.7386	count
ring_length_r30_px	headline	270.778	255	109.167	px
ring_length_r60_px	reported	847.222	815	240.089	px
owned_length_px	ownership-dependent	2723.75	2439.16	1430.97	px
edge_count	ownership-dependent	112.889	90	58.5543	count
mean_edge_length_px	ownership-dependent	24.138	24.1478	2.3011	px
perimeter_px	normalised	236.382	238.108	53.1676	px
ring_area_r30_px2	normalised	9046.22	8595	1273.77	px^2
ring_area_r60_px2	normalised	22951.3	21925	2828.56	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9957	1	0.0128	unitless
ring_density_r30	normalised	0.0293234	0.0281559	0.00914184	px^-1
ring_density_r60	normalised	0.0366306	0.0375317	0.007931	px^-1
roots_per_100px_perimeter	normalised	2.8471	2.7514	1.0992	per 100 px
ring_attached_length_r30_px	network v2	203	189	116.572	px
ring_attached_length_r60_px	network v2	540.556	568	266.612	px
ring_length_w_r30_px	network v2	306.653	292.35	127.837	px
ring_length_w_r60_px	network v2	964.439	937.749	276.295	px
sholl_crossings_r10	network v2	6.4444	7	3.3208	count
sholl_crossings_r20	network v2	10.7778	11	3.5629	count
sholl_crossings_r30	network v2	13.2222	13	4.0552	count
ring_width_mean_r30_px	width	6.4086	6.4765	0.4389	px

Per field
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measure	value	units
analysed_area_px2	1.02836e+06	px^2
total_skeleton_length_px	35500	px
canalicular_length_density_per_px	0.034521	px^-1
junction_count	786	count
junction_density_per_px2	0.00076432	px^-2
lacuna_count	10	count
lacunae_per_px2	9.72e-06	px^-2
skeleton_component_count	192	count

mean_component_length_px	184.896	px
median_component_length_px	20	px
field_length_density_w_per_px	0.0392772	px^-1
field_density_without_flagged_per_px	0.034521	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.0431	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	35500	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2174	80.8669	35.2412	2.2947	0.9	0.9518	0.4109	79.3965	270.898	False
2	3447	90.3677	63.9486	1.4131	0.7066	0.674	0.4223	197.059	916.902	False
3	1883	69.2617	34.8694	1.9863	0.864	0.9448	0.5742	250.833	340.47	False
4	2052	69.2015	39.4228	1.7554	0.8219	0.9112	0.0622	388.671	369.939	False
5	2266	88.1613	35.5691	2.4786	0.915	0.9021	0.1324	473.05	745.191	False
6	1344	87.6934	23.0299	3.8078	0.9649	0.7764	0.2206	545.19	876.866	False
7	1683	71.0123	31.3735	2.2634	0.8971	0.9355	-0.0518	634.68	198.315	False
8	2203	91.2092	31.4053	2.9043	0.9389	0.9295	0.0832	734.301	635.44	False
9	2633	93.5592	36.2778	2.579	0.9218	0.9454	0.0457	780.576	481.495	False
10	531	28.7881	24.8252	1.1596	0.5063	0.9332	0.5942	1012.28	184.834	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	4	242	815	1024.43	42	24.3912	5.9711
2	10	471	1283	2439.16	107	22.7959	6.4673
3	8	294	899	4168.65	189	22.0564	6.8444
4	7	228	571	4915.99	188	26.1489	5.9029
5	11	339	1012	4467.34	185	24.1478	6.975
6	3	162	771	1688.54	79	21.3739	5.7543
7	4	99	493	1398.4	64	21.85	6.5063
8	6	347	1011	1967.06	72	27.3203	6.4765
9	7	255	770	2444.15	90	27.1572	6.7796
10	1	79	410	5.8284	1	5.8284	6.0343

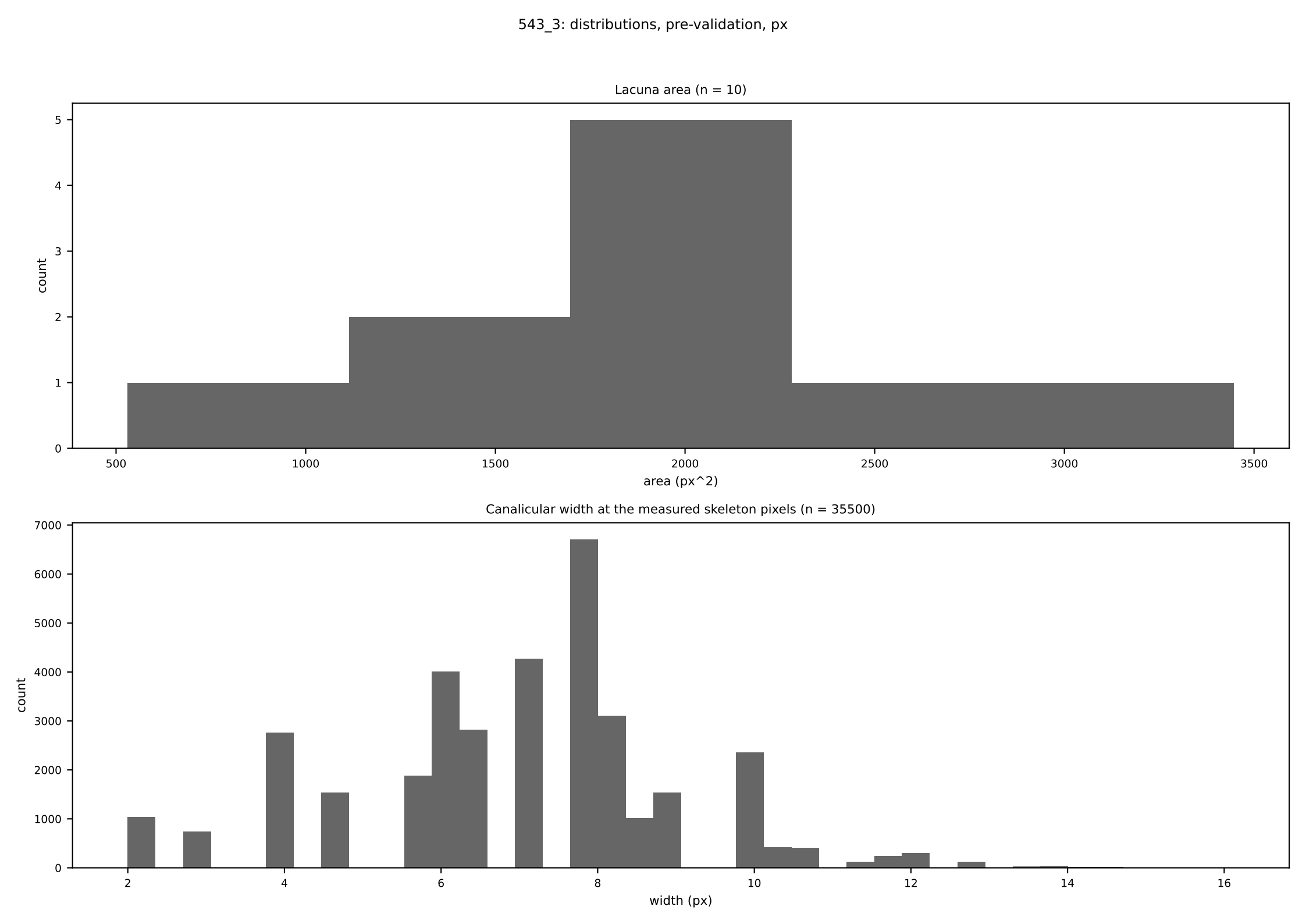
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	210.066	8595	21715	1	0.9616	0.0281559	0.0375317	1.9042
2	363.446	12097	29071	1	1	0.0389353	0.0441333	2.7514
3	184.794	7941	19515	1	1	0.037023	0.0460671	4.3291

4	205.965	8333	20507	1	1	0.0273611	0.0278442	3.3986
5	241.907	9380	23807	1	1	0.0361407	0.0425085	4.5472
6	248.25	8529	21925	1	1	0.018994	0.0351653	1.2085
7	191.965	7943	21525	1	1	0.0124638	0.0229036	2.0837
8	238.108	9212	24077	1	1	0.0376683	0.0419903	2.5199
9	242.936	9386	24420	1	1	0.0271681	0.0315315	2.8814
10	95.3909	3354	9525	0.6102	0.5725	0.023554	0.0430446	1.0483

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	114	322	279.25	937.749	4	11	15
2	421	914	547.044	1460.11	12	16	18
3	243	764	323.794	1029.89	8	11	18
4	189	505	250.238	646.772	7	10	9
5	295	746	378.907	1158.39	9	12	13
6	68	210	182.681	874.281	2	7	11
7	57	146	107.669	550.789	2	4	6
8	258	690	397.948	1148.69	8	14	16
9	182	568	292.35	873.281	6	12	13
10	6	6	84.3553	467.061	1	3	7


543_3: distributions, pre-validation, px
Lacuna area (n = 10)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 35500)
count
width (px)